Sparse-Concept Calibration of Knowledge Tracing Models for Threshold-Based Educational Decisions

Khanh-Trinh Nguyen[1], Tuan Dao Minh[1], Duong Nguyen Tien[1], Chi Thanh Nguyen[2], and Van-Hau Nguyen[1,*]
1 Hung Yen University of Technology and Education, Hung Yen, Vietnam
2 Academy of Military Science and Technology, Ha Noi, Vietnam

Email: trinhnk@utehy.edu.vn (K.-T.N.); tuanymc@utehy.edu.vn (T.D.M.); duongnt@utehy.edu.vn (D.N.T.); thanhnc@ioit.ai.vn (C.T.N.); haunv@utehy.edu.vn (V.-H.N.)
*Corresponding author

Abstract—*Knowledge Tracing (KT) systems that skip, remediate, or advance practice typically consume a predicted probability rather than an area-under-the-curve (AUC) score, so population ranking can look acceptable while probabilities are poorly calibrated on rarely practiced knowledge components (KCs). This paper reports a diagnostic evaluation—not a new KT architecture—of train-only KC-frequency strata on three public logs (ASSISTments 2012, Junyi Academy, and XES3G5M) with IRT, DKT, and SimpleKT. Lower KC training frequency does not universally degrade discrimination, but calibration can become less reliable in some sparse-concept regimes. On ASSISTments 2012, SimpleKT expected calibration error (ECE) rises from 0.114 on dense KCs to 0.228 on sparse KCs (Limited occupancy, N≈415); Junyi has no learner-based sparse stratum, and XES3G5M SimpleKT ECE is essentially flat. A locked global threshold at τ=0.7 raises the SimpleKT false-advance rate (incorrect-response rate among advance decisions) from 0.196 (dense) to 0.268 (sparse) on one ASSISTments fold; the sparse–dense gap stays positive on all five training seeds (mean 0.047; four unique student partitions). This is a simulated decision gate, not a classroom intervention. Probability-threshold decisions should be validated by KC-frequency stratum rather than on population AUC or ECE alone.*

Keywords—*knowledge tracing, calibration, sparse concepts, learning analytics, educational decision support, mastery threshold*

I. Introduction

Contributions are conservative: (i) a train-only KC-frequency protocol with an explicit strict cold-start group, so the definition of "sparse" cannot leak test-fold counts; (ii) per-stratum calibration (ECE and Brier decomposition) on three public datasets, with occupancy flags (Reliable / Limited / Insufficient); (iii) a locked-threshold simulation of FAR and miss rates, with a five-seed check of the sparse–dense FAR gap on ASSISTments 2012. We do not propose a new KT architecture, a new calibration algorithm, or a new auditing theory, and we do not report a classroom intervention. A graph-KT (GKT) model and a CL4KT-style adapter appear only as an exploratory single-fold diagnostic on ASSISTments.

The empirical answers are bounded. Lower KC training frequency does not universally degrade discrimination: on XES3G5M, sparse AUC is higher than dense AUC for DKT and SimpleKT. Calibration can, however, become less reliable in some sparse-concept regimes. On ASSISTments 2012, SimpleKT expected calibration error (ECE) increases from dense to sparse KCs, and a locked gate at τ=0.7 yields a higher FAR on sparse than on dense advances. The same ECE gradient is absent on Junyi, where the learner-based sparse stratum is empty, and is essentially absent for SimpleKT on XES3G5M. We therefore do not claim that sparse KCs always fail, and we do not claim a causal effect of training frequency on calibration.

This paper treats that mismatch as an evaluation problem for educational technology, not as a reason to propose another KT architecture. The study is organized around three research questions. RQ1: Does lower KC training frequency systematically degrade predictive discrimination? RQ2: How does calibration vary across KC-frequency strata and datasets? RQ3: When a fixed probability threshold is applied, does decision-error behavior differ between sparse and dense KCs?

KT evaluations typically report population AUC and accuracy [4], [5]. Those metrics rank models, but they can conceal miscalibration on low-frequency KCs—skills with little training-fold evidence—because most test events lie on dense, frequently practiced concepts. A model can discriminate well in aggregate and still assign overconfident probabilities on the sparse tail. When a fixed threshold is applied to p, that overconfidence appears as an advance decision followed by an incorrect next response. We term this the false-advance rate (FAR). Section III defines FAR as $P(y=0 \mid p \geq \tau)$; y denotes observed next-response correctness, not latent mastery truth.

Adaptive practice platforms and intelligent tutoring systems use Knowledge Tracing (KT) to estimate a learner's command of a skill and to decide whether to skip, remediate, or advance practice [1]–[3]. Those systems rarely consume a population area-under-the-curve (AUC) statistic. They consume a predicted probability p, which is then compared with a threshold τ. If p≥τ, the platform typically withholds additional practice on that skill. The operational chain is therefore population AUC → predicted probability → calibration → threshold-based educational decisions, with particular diagnostic risk on sparsely trained knowledge components (KCs).

II. Literature Review

Classical KT uses Bayesian knowledge tracing or item-response models [1], [6]. Deep sequential models (DKT, attentive and transformer KT) improved aggregate prediction [2], [4], [7]. Graph and contrastive variants often motivate architecture by sparse or related skills [8], [9]. Those papers still report mainly population metrics.

Educational measurement has long warned that ranking is not reliability [10]. Calibration error and reliability diagrams quantify whether predicted probabilities match empirical correctness [11]–[14]. Recent KT work has begun to study calibration and sparse-concept evaluation [15]–[17], but deployed tutoring systems still typically consume a single global p. This study sits in that gap: we keep standard models, stratify by training-only frequency, and ask whether a population gate would treat sparse skills more leniently—or more dangerously—than dense ones.

III. Materials and Methods

A. Datasets, Splits, and Models

We use three de-identified public logs after a common schema: ASSISTments 2012 [18], [5], Junyi Academy [19], and XES3G5M [20]. Table 1 summarizes processed cohort size. Learner-based splits (unseen students) are the primary setting. Temporal splits are a complementary stress test and are not the source of the gate numbers below.

TABLE 1. PROCESSED COHORT STATISTICS (LEARNER-BASED SPLIT)

Dataset	Learners	KCs	Interactions	Test events
ASSISTments 2012	27,806	265	2.66M	534,150
Junyi Academy	71,014	1,326	16.2M	3,269,022
XES3G5M	18,066	866	7.95M	1,589,145

Baselines: one-parameter IRT [6] as a classical reference, DKT [2], and SimpleKT [4]. On ASSISTments fold 0 only we also score a train-only GKT graph [8] and a CL4KT protocol adapter [9] (contrastive views on training sequences; not an official CL4KT checkpoint). IRT under learner-based splits has no ability parameter for unseen students, so its AUC is 0.50 by construction; we report it as a base-rate reference, not as a ranking competitor.

Deep models are trained at seeds 42, 2024, 2025, 2026, and 2027. The processed partitions for seeds 2025 and 2026 are identical (fold_2 = fold_3) on all three datasets. Tables that report mean±sd therefore summarize four unique student partitions: the two initializations on the duplicated split are averaged first. The gate-robustness table still lists all five training seeds without dropping one after the fact.

B. Train-Only Frequency Strata

For each fold, KC frequency f_train is counted on the training file only. Buckets: strict cold-start ($f=0$), very sparse ($0<f<20$), sparse ($20\leq f<100$), medium ($100\leq f<500$), dense ($f\geq 500$). Dense KCs dominate event volume while a non-empty sparse-like tail exists on ASSISTments and XES3G5M. Occupancy flags follow test-event count N: Reliable ($N\geq 1000$), Limited ($100\leq N<1000$), Insufficient ($N<100$). Success claims require Limited or Reliable occupancy. Very-sparse ASSISTments cells are Insufficient and descriptive only.

Fig. 1. Train-only KC-frequency strata. Dense concepts dominate interactions; sparse and cold-start KCs remain the diagnostic tail.

C. Calibration

Let $y\in\{0,1\}$ be correctness and $p\in[0,1]$ the predicted probability. With M=15 equal-width bins, ECE = Σ_m (n_m / N) |acc_m − conf_m|. We also report the Brier score and its reliability / resolution / uncertainty decomposition [14]. Lower ECE is better calibration; it is not a substitute for AUC.

D. Simulated Mastery Gate

For a locked global threshold τ, the system advances (skips remediation) if $p\geq\tau$ and triggers remediation otherwise. Sparse events never receive their own tuned τ. The display threshold is τ=0.7; the grid {0.5, 0.6, 0.7, 0.8} is recorded in the artifact and is not a search over sparse error.

Let $A=\{p\geq\tau\}$. False mastery (FM) is $P(y=0 \mid A)$: among advances, how often was the answer wrong. Miss is $P(A \mid y=0)$: among wrong answers, how often the system still skipped help. If the model were calibrated on the advance set, FM would match $E[1-p \mid A]$. We report ΔFM = FM_sparse − FM_dense. Positive ΔFM means sparse advances are dirtier than dense advances. This is a simulated decision error, not an instructional RCT.

IV. Result and Discussion

A. Aggregate Ranking Is Not the Sparse Story

Table 2 shows learner-based AUC. DKT is slightly above SimpleKT on all three datasets. IRT AUC is 0.50 for the reason given above. These numbers would look like a routine KT bake-off. They do not tell a designer whether p is trustworthy on the tail.

TABLE 2. OVERALL LEARNER-BASED AUC (MEAN±SD OVER FOUR UNIQUE PARTITIONS)

Dataset	Model	AUC	ACC
ASSISTments 2012	IRT	0.5000	0.6973±0.0004
ASSISTments 2012	DKT	0.6979±0.0014	0.7182±0.0014
ASSISTments 2012	SimpleKT	0.6837±0.0025	0.6996±0.0032
Junyi Academy	IRT	0.5000	0.7053±0.0018
Junyi Academy	DKT	0.7320±0.0009	0.7343±0.0015
Junyi Academy	SimpleKT	0.7231±0.0030	0.7274±0.0015
XES3G5M	IRT	0.5000	0.7961±0.0031
XES3G5M	DKT	0.8171±0.0022	0.8327±0.0032
XES3G5M	SimpleKT	0.7557±0.0013	0.8067±0.0037

On XES3G5M, sparse AUC is higher than dense AUC (DKT 0.857 vs. 0.817; SimpleKT 0.847 vs. 0.755; Reliable occupancy). Lower frequency is therefore not a universal ranking failure. Junyi's learner-based sparse bucket is empty under the pre-registered cuts—a protocol outcome, not a missing cell to impute.

B. Calibration Can Move When Ranking Does Not

Table 3 is the calibration punchline. On ASSISTments 2012, SimpleKT ECE increases from 0.1136±0.0066 (dense, Reliable, N=523,971) to 0.1541 (medium) to 0.2280±0.0197 (sparse, Limited, N=415). DKT shows the same direction (dense ECE 0.060 vs. sparse 0.233). IRT dense ECE is tiny (0.003) but resolution is zero: a constant-like predictor can look "calibrated" while ranking nothing.

On Junyi, only dense and medium strata exist; SimpleKT ECE rises modestly from 0.079 to 0.107. On XES3G5M, SimpleKT ECE is essentially flat (0.114→0.111→0.125 dense/medium/sparse) despite Reliable sparse occupancy (N=2,010). ECE-flat is not a license to skip occupancy reporting, and—as the gate results show—it is not the same as a flat miss rate.

TABLE 3. SIMPLEKT EVENT-LEVEL ECE BY TRAIN-ONLY FREQUENCY STRATUM (FOUR UNIQUE PARTITIONS). FLAGS: R RELIABLE, L LIMITED. JUNYI SPARSE IS EMPTY.

Dataset	Stratum	N	Flag	ECE
ASSISTments 2012	dense	523,971	R	0.1136±0.0066
ASSISTments 2012	medium	5,963	R	0.1541±0.0051
ASSISTments 2012	sparse	415	L	0.2280±0.0197
Junyi Academy	dense	3,232,614	R	0.0792±0.0051
Junyi Academy	medium	3,836	R	0.1073±0.0156

XES3G5M	dense	1,268,696	R	0.1145±0.0011
XES3G5M	medium	12,980	R	0.1114±0.0076
XES3G5M	sparse	2,010	R	0.1248±0.0085

C. A Global Gate Can Turn the ASSISTments Gradient into a Decision Error

Table 4 applies τ=0.7 on ASSISTments fold 0 (seed 42; sparse N=444, Limited). SimpleKT false mastery among advances is 0.196 on dense KCs and 0.268 on sparse KCs (ΔFM=+0.072). The calibrated expectation E[FM] on sparse advances is only 0.050, so the excess error is large. DKT shows a similar FM gap. Train-only GKT shrinks ΔFM to +0.015 (FM 0.205 → 0.220). The CL4KT adapter is intermediate (0.185→0.240). We do not read this as "use GKT in production"; we read it as: the ASSISTments calibration gradient is not a property of the dataset alone.

TABLE 4. SIMULATED GATE AT T=0.7, ASSISTMENTS 2012 FOLD 0. ADVANCE IF P≥T. NOT A CLASSROOM TRIAL.

Model	Stratum	N	FM	E[FM]	Miss
SimpleKT	dense	528,018	0.196	0.113	0.352
SimpleKT	sparse	444	0.268	0.050	0.320
DKT	dense	528,018	0.200	0.147	0.398
DKT	sparse	444	0.296	0.057	0.365
GKT (train-only)	dense	528,018	0.205	0.163	0.455
GKT (train-only)	sparse	444	0.220	0.157	0.234
CL4KT (adapter)	dense	528,018	0.185	0.175	0.359
CL4KT (adapter)	sparse	444	0.240	0.116	0.244

Table 5 checks whether SimpleKT ΔFM is a one-fold accident. It is positive on all five training seeds (mean 0.047, sd 0.033). A KC-clustered bootstrap on seed 42 yields a 95% interval [0.006, 0.138]—excluding 0, but wide, as Limited occupancy requires. DKT ΔFM is positive on only three of five seeds and is not treated as a five-run finding. On XES3G5M, SimpleKT ΔFM is negative on all five seeds, but ΔMiss is positive on all five (mean +0.112): among actual incorrect answers, the system still advances more often on sparse than on dense KCs. A flat ECE therefore does not imply a flat miss rate.

Coincidence of digits: SimpleKT dense E[FM] at τ=0.7 on seed 42 is 0.113, close to the four-partition dense ECE 0.114. They are different quantities.

TABLE 5. GATE ROBUSTNESS AT T=0.7 ON ASSISTMENTS 2012 (FIVE TRAINING SEEDS; FOUR UNIQUE PARTITIONS). GKT/CL4KT REMAIN SEED 42 ONLY.

Model	Mean ΔFM	SD	Seeds >0	Mean sparse N
SimpleKT	0.047	0.033	5/5	413
DKT	0.033	0.048	3/5	413

D. What a Designer Should Take Away

If a learning platform uses KT probabilities only to sort students, Table 2 may suffice. If it uses p to skip practice or declare mastery, Table 2 is the wrong dashboard. The ASSISTments SimpleKT cell is the cautionary case: competitive AUC, Limited-but-estimable sparse occupancy (about 19% of KCs fall below the sparse threshold on fold 0), a monotonic ECE rise, and a gate that advances more dirty sparse attempts. The XES3G5M cell is the opposite caution: plenty of sparse mass and Reliable occupancy, yet SimpleKT ECE does not degrade—while miss rates still can. Junyi shows that the protocol can refuse a sparse claim when the bucket is empty.

Three observable conditions track when sparse-calibration claims are informative rather than obligatory (Table 6): (1) a non-empty low-frequency tail under the split actually used; (2) enough sparse test events for at least a Limited-flag ECE; (3) frequency–difficulty coupling that is not inverted. ASSISTments meets all three. That is a diagnostic pre-condition, not a claim that training frequency causes miscalibration.

TABLE 6. WHEN A SPARSE-CALIBRATION CLAIM IS ESTIMABLE (FINDINGS, NOT CAUSES). SPARSE MASS: SHARE OF KCS WITH F_TRAIN<100.

Condition	ASSISTments	Junyi	XES3G5M
Sparse mass	18.9% of KCs	0%	22.5% of KCs
Sparse N	415 (L)	empty	2,010 (R)
Difficulty coupling	ρ=−0.227 (sparse harder)	ρ=−0.416	ρ=+0.087 (weak, inverted)
SimpleKT ECE	0.114→0.228	dense→medium only	flat 0.114→ 0.125

E. What This Paper Does Not Show

The gate is not a classroom policy and not an A/B test of remediation. GKT and the CL4KT adapter are single-fold, ASSISTments-only instantiations. Temporal results are a single corrected cutoff (seed 42), not a multi-cutoff variance estimate. IRT is not a fair ranking baseline on unseen learners. We did not tune τ on sparse events, and we do not recommend doing so in deployment without a separate validation design.

F. Implications for Information and Education Technology

IJIET's audience includes operators of educational platforms, not only KT researchers. For that audience, the operational recommendation is narrow: (i) log train-only KC frequency with every prediction; (ii) report ECE and occupancy on dense vs. sparse slices before enabling a global mastery threshold; (iii) if a threshold must be global, simulate FM and Miss on the sparse slice at the same τ used in the product; (iv) do not treat a population AUC win, or a graph/contrastive architecture, as automatically safer on the tail.

V. CONCLUSION

KT models that look adequate on AUC can still be poorly calibrated on rarely practiced skills, and a global mastery threshold can then produce a higher false-mastery rate on those skills. That pattern appears for SimpleKT on ASSISTments 2012 (ECE 0.114→0.228, Limited N=415; seed-42 FM 0.196→0.268; ΔFM>0 on 5/5 seeds). It does not appear as a universal law: Junyi has no learner-based sparse stratum, and XES3G5M SimpleKT ECE is flat. Sparse-concept diagnostics are therefore conditionally important. The simulation is a decision-error check for educational-technology systems, not a claim of a new KT model and not a classroom intervention.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

K.-T.N. led the diagnostic design, experiments, and manuscript. T.D.M. and D.N.T. contributed data processing and baseline runs. C.T.N. contributed methodological review. V.-H.N. supervised the study and revised the

manuscript. All authors approved the final version.

ACKNOWLEDGMENT

Anonymized code and prediction exports are available for peer review; a public repository will be released upon acceptance. This study uses de-identified public learner logs (ASSISTments 2012, Junyi Academy, XES3G5M) and is not a classroom trial.

REFERENCES

[1] A. T. Corbett and J. R. Anderson, "Knowledge Tracing: Modeling the acquisition of procedural knowledge," User Modeling and User-Adapted Interaction, vol. 4, no. 4, pp. 253–278, 1994.

[2] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L. J. Guibas, and J. Sohl-Dickstein, "Deep Knowledge Tracing," in Advances in Neural Information Processing Systems, 2015, pp. 505–513.

[3] G. Abdelrahman, Q. Wang, and B. Nunes, "Knowledge Tracing: A survey," ACM Computing Surveys, vol. 55, no. 11, pp. 1–37, 2023.

[4] Z. Liu, Q. Liu, J. Chen, S. Huang, and W. Luo, "SimpleKT: A simple but tough-to-beat baseline for Knowledge Tracing," in The Eleventh International Conference on Learning Representations, 2023.

[5] Z. Liu, Q. Liu, J. Chen, S. Huang, J. Tang, and W. Luo, "pyKT: A python library to benchmark deep learning based Knowledge Tracing models," in Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2022.

[6] G. Rasch, Probabilistic Models for Some Intelligence and Attainment Tests. Danish Institute for Educational Research, 1960.

[7] A. Ghosh, N. Heffernan, and A. S. Lan, "Context-aware attentive Knowledge Tracing," in Proc. ACM SIGKDD Conf. Knowledge Discovery and Data Mining, 2020, pp. 2330–2339.

[8] H. Nakagawa, Y. Iwasawa, and Y. Matsuo, "Graph-based Knowledge Tracing: Modeling student proficiency using graph neural network," in IEEE/WIC/ACM International Conference on Web Intelligence, 2019, pp. 156–163.

[9] W. Lee, J. Chun, Y. Lee, K. Park, and D. Choi, "Contrastive learning for Knowledge Tracing," in Proc. ACM Web Conference, 2022, pp. 2330–2338.

[10] R. Pelánek, "Metrics for evaluation of student models," Journal of Educational Data Mining, vol. 7, no. 2, pp. 1–19, 2015.

[11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proc. International Conference on Machine Learning, 2017, pp. 1321–1330.

[12] M. P. Naeini, G. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in Proc. AAAI Conference on Artificial Intelligence, vol. 29, no. 1, 2015.

[13] G. W. Brier, "Verification of forecasts expressed in terms of probability," Monthly Weather Review, vol. 78, no. 1, pp. 1–3, 1950.

[14] M. H. DeGroot and S. E. Fienberg, "The comparison and evaluation of forecasters," The Statistician, vol. 32, no. 1/2, pp. 12–22, 1983.

[15] X. Yan, C. Tang, and A. Shimada, "Recovering stranded discrimination in Knowledge Tracing: Per-item bias correction via empirical-Bayes shrinkage," arXiv:2606.14123, 2026.

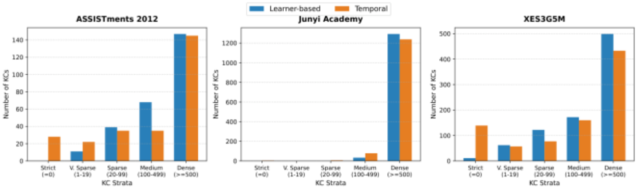
[16] S. Huang, Z. Liu, X. Zhao, W. Luo, and J. Weng, "Towards robust Knowledge Tracing models via k-sparse attention," in Proc. 46th International ACM SIGIR Conference, 2023, pp. 2441–2445.

[17] I. Bhattacharjee and C. Wayllace, "Cold start problem: An experimental study of Knowledge Tracing models with new students," in Artificial Intelligence in Education (AIED 2025), LNCS vol. 15880, Springer, 2025, pp. 425–432.

[18] ASSISTmentsData, "ASSISTments 2012–2013 school data with affect," ASSISTments Public Datasets, 2012. [Online]. Available: https://sites.google.com/site/assistmentsdata/datasets/2012-13-school-data-with-affect

[19] Junyi Academy, "Junyi Academy online learning activity dataset," Kaggle, 2019. [Online]. Available: https://www.kaggle.com/datasets/junyiacademy/learning-activity-public-dataset-by-junyi-academy

[20] Z. Liu, Q. Liu, T. Guo, J. Chen, S. Huang, X. Zhao, J. Tang, W. Luo, and J. Weng, "XES3G5M: A Knowledge Tracing benchmark dataset with auxiliary information," in Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2023.



Copyright © 2026 by the authors. This is an open access article distributed under the Creative Commons Attribution License which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited (CC BY 4.0).